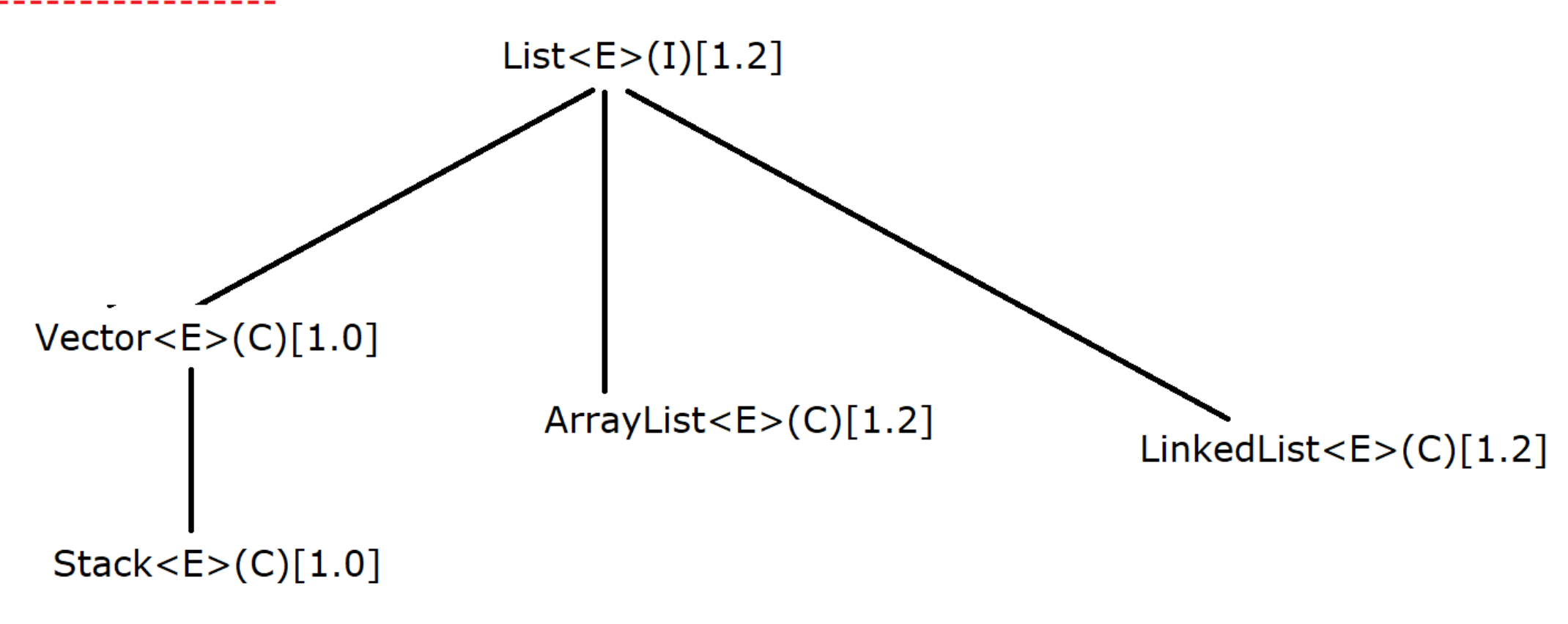
List<E> interface :

- It is the sub interface of Collection interface available from JDK 1.2V.
- It accepts duplicate objects.
- It stores the object based on the index position.
- It can accept homogeneous, hetreogeneous and null elements.
- We can directly perform sorting operation on List interface implemented classes using Collections.sort().

List<E> interface Hierarchy :



Common behavior of List interface implemented class :
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1) It stores the elements on the basis of index because internally it is using array concept.

2) It can accept duplicate, homogeneous and heterogeneous elements.

3) It stores everything in the form of Object.

Example :

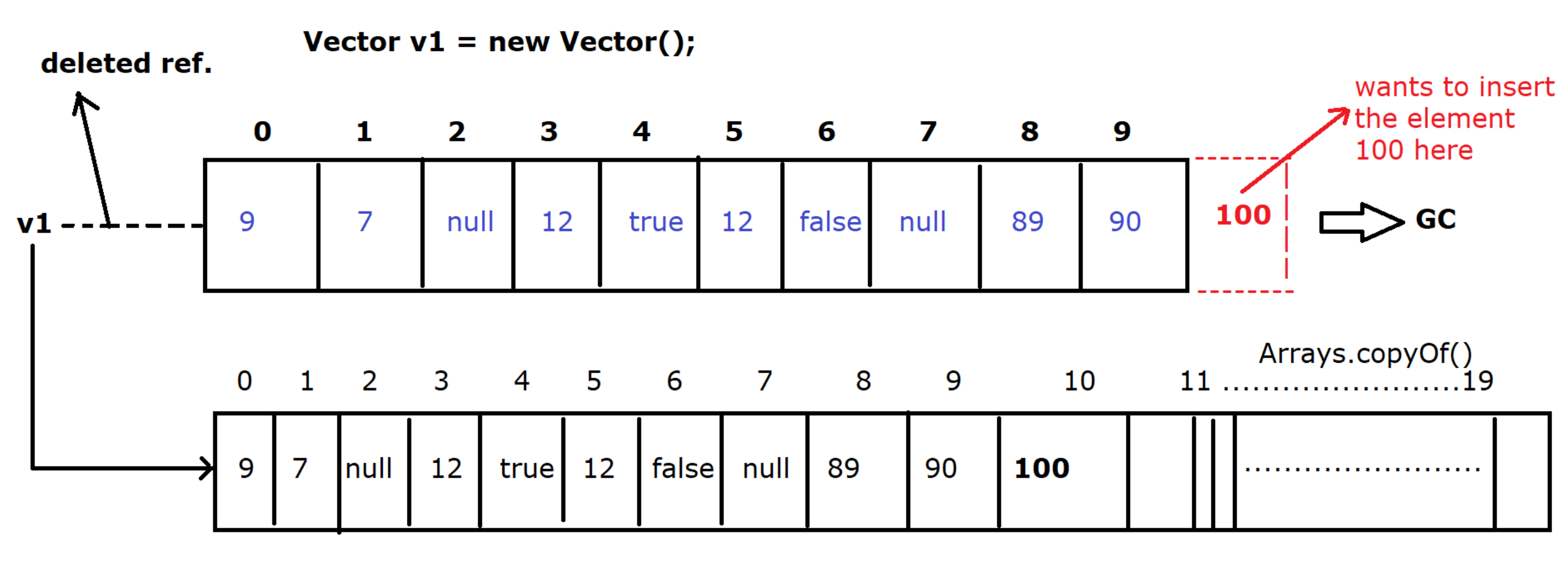
```
List list = new Vector();
 list.add(12);
 list.add(13);
 list.add(14);
 list.add(15);
 list.add(16);

 for(Object i : list)
 {
      IO.println(i);
 }
```

4) When we accept the collection classes without generic concept then compiler generates a warning message because It is unsafe object.

5) By using generic (<>) (JDK 1.5V) we can eliminate compilation warning and still we can take homogeneous as well as hetrogeneous.(Vector<Object>)

6) In list interface few classes are dynamically Growable like **Vector and ArrayList**

Vector v1 = new Vector();



List<E> interface Methods
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1) public boolean isEmpty() :- Verify whether List is empty or not

2) public void clear() :- Will clear all the elements, Basically List will become empty.

3) public int size() :- To get the size of the Collections(Total number of elements are available in the collection)

4) public void add(int index, E element) :- Insert the element based on the index position.

5) public boolean addAll(int index, Collection c) :- Insert the Collection based on the index position

6) public E get(int index) :- To retrieve the element based on the index position.

7) public E set(int index, E element) :- To override or replace the existing element based on the index position.

8) public E remove(int index) :- remove the element based on the index position

9) public boolean remove(Object element) :- remove the element based on the object element, It is the Collection interface method extended by List interface implemented by Vector class.

10) public int indexOf(Object obj) :- index position of the element otherwise -1

11) public int lastIndexOf(Object obj) :- last index position of the element

12) public Iterator iterator() :- To fetch or iterate or retrieve the elements from Collection in forward direction only.

13) public ListIterator listIterator() :- To fetch or iterate or retrieve the elements from Collection in forward and backward direction.